ECE 471 Formula Sheet

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Multiple Interference

$$r_f = \frac{E_r}{E_i} = \frac{r_{12} + r_{23}e^{-j\phi}}{1 - r_{21}r_{23}e^{-j\phi}} \qquad t_f = \frac{E_t}{E_i} = \frac{t_{12}t_{23}e^{-j\phi/2}}{1 - r_{21}r_{23}e^{-j\phi}}$$

$$\phi = 2n_2k_0d \qquad R_f = |r_f|^2 \qquad T_f = \frac{n_3}{n_1}|t_f|^2 \qquad R_f + T_f = 1$$

$$A = \frac{t_{12}E_i}{1 - r_{21}r_{23}e^{-j\phi}} \qquad r_{from,to},\ t_{from,to}$$

Anti-reflection Coating

Set $\phi = 2n_2k_0d = m\pi$ with m odd

$$R_{f,min} = \left(\frac{n_2^2 - n_1n_3}{n_2^2 + n_1n_3}\right) \implies \text{Set } n_2 = \sqrt{n_1n_3} \text{ for } R_{f,min} = 0$$

Fabry-Perot Resonator

$$\phi = n\left(\frac{2\pi}{\lambda}\right)2L = 2m\pi \qquad \lambda_m = \frac{2nL}{m} \qquad \nu_m = \frac{c}{\lambda_m} = \frac{mc}{2nL}$$

$$\Delta\nu_{FSR} = \nu_{m+1} - \nu_m = \frac{c}{2nL} \qquad \delta\nu_m = \frac{\Delta\nu_{FSR}}{F} \qquad R = r^2$$

$$F = \frac{\pi\sqrt{R}}{1-R} \qquad Q = \frac{\nu_m}{\delta\nu_m} = mF \qquad I = \frac{I_0}{(1-R)^2 + 4R\sin^2(\phi/2)}$$

$$\frac{I_t}{I_i} = \frac{(1-R)^2}{(1-R)^2 + 4R\sin^2(\phi/2)} \qquad \frac{I_r}{I_i} = 1 - \frac{I_t}{I_i}$$

Dielectric Mirrors

$$d_1 = \frac{\lambda_1}{4} = \frac{\lambda_0}{4n_1} \quad d_2 = \frac{\lambda_2}{4} = \frac{\lambda_0}{4n_2} \quad R_N = \left[\frac{n_1^{2N} - (n_0/n_3)n_2^{2N}}{n_1^{2N} + (n_0/n_3)n_2^{2N}}\right]^2$$

$$\frac{\Delta\lambda}{\lambda_o} \approx \frac{4}{\pi}\sin^{-1}\left(\frac{n_1 - n_2}{n_1 + n_2}\right) \qquad n_0 < n_1,\ n_1 > n_2$$

Fraunhofer Diffraction

Linear lit Diffraction

$$I(\theta) = I_0\,\mathrm{sinc}^2(\beta) \qquad \beta = \frac{1}{2}ka\sin(\theta) = \frac{1}{2}k_ya$$

$$sin(\theta_m) = \frac{m\lambda}{a} \qquad sin(\theta_D) = \frac{\lambda}{a} \qquad \Delta\theta = 2\theta_D = \frac{2\lambda}{a}$$

Circular Aperture Diffraction

$$sin(\theta_D) = 1.22\frac{\lambda}{D} \text{ Rayleigh criterion } \implies sin(\Delta\theta_{min}) = 1.22\frac{\lambda}{D}$$

Diffraction Gratings

Transmission Grating

$$d\sin(\theta_m) = m\lambda \qquad I(y) = I_o\,\mathrm{sinc}^2\left(\frac{1}{2}k_ya\right)\left[\frac{\sin\left(\frac{1}{2}Nk_yd\right)}{N\sin\left(\frac{1}{2}k_yd\right)}\right]$$

$$k_y = \frac{2\pi}{\lambda}\sin(\theta) = \frac{2\pi}{\lambda}\frac{y}{L}$$

Reflection Grating

$$d\sin(\theta_m) - d\sin(\theta_i) = m\lambda$$

Gaussian Beams

$$w_0 = \sqrt{\frac{2z_0}{\pi}} = \sqrt{\frac{\lambda z_0}{\pi}} \qquad R(z) = z\left[1 + \left(\frac{z_0}{z}\right)^2\right] \qquad \theta = \frac{w_0}{z_0} = \frac{\lambda}{\pi w_0}$$

$$q(z) = z + jz_0 \implies \frac{1}{q(z)} = \frac{1}{R(z)} - j\frac{w_0^2}{z_0w^2(z)} = \frac{1}{R(z)} - j\frac{\lambda}{\pi w^2(z)}$$

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D}$$

Models of Light Interaction

Lorentz Model

$$\epsilon_r = \epsilon_r' - j\epsilon_r'' = 1 + \chi = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 + j\Gamma\omega} = \tilde{n}^2 \qquad \tilde{n} = n - j\kappa$$

$$E(z,t) = E_0e^{j(\omega t - nk_0z)}e^{-\kappa k_0z} \qquad \alpha = 2\kappa k_0 \qquad I = I_0e^{-\alpha z}$$

Drude Model

$$\omega_0 = 0 \qquad \epsilon_r = 1 - \frac{\omega_p^2}{\omega^2 - j\Gamma\omega} = \tilde{n}^2 \qquad \text{Ngl. Damping} \implies \Gamma = 0$$

Optics of Crystals

Ord.: $(E_y)\ k = n_o\frac{\omega}{c} \quad v = \frac{c}{n_o}$ Extr.: $(E_z)\ k = n_e\frac{\omega}{c} \quad v = \frac{c}{n_e}$

QWP $\phi_e - \phi_0 = \pi/2$ HWP $\phi_e - \phi_0 = \pi$

$$\vec{E_{out}} = \left(E_ye^{-j\phi_o},\ E_ze^{-j\phi_e}\right)$$

Photodetectors

$$I_{ph} = RP_0 \qquad i_D = I_0\left(e^{qv_D/\eta k_BT} - 1\right) \qquad i = i_d - I_{ph}$$

$$\eta_{int} = \frac{EHP\ gen.}{photon\ abs.} \qquad \eta_{ext} = \frac{h\nu}{e}\frac{I_{ph}}{P_0} = \frac{Electron\ flow}{Photon\ flow} \qquad R = \frac{I_{ph}}{P_0}$$

Light Sources

$$M_\lambda(\lambda) = \frac{3.741\cdot10^8}{\lambda^5}\frac{1}{e^{(hc/\lambda k_BT)} - 1}\ [W/m^2\cdot\mu m] \qquad \lambda_{pk} = \frac{2900}{T}$$

Light Emitting Diodes

$$E_{ph} = h\nu = E_2 - E_1 \qquad E_{pk} = E_g + \frac{1}{2}k_BT \qquad \Delta E_{ph} \approx mk_VT$$

$$\eta_{EQE} = \frac{e}{h\nu}\frac{P_0}{I} \qquad \eta_{PCE} = \frac{P_0}{IV_D}$$

Lasers

$$R_{ab} = B_{12}N_1\rho(\nu) \qquad R_{sp} = A_{21}N_2 \qquad R_{st} = B_{21}N_2\rho(\nu)$$

$$B_{21} = B_{12} \qquad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \qquad \tau_{sp} = 1/A_{21}$$

$$\frac{dN_{ph}}{dt} = B_{21}(N_2 - N_1)\rho(\nu) \qquad \alpha = \sigma_{ab}N_1 \qquad g = \sigma_{em}N_2$$

$$\frac{dI}{dx} = g_{net}I \qquad g_{net} = g - \alpha \approx \frac{B_{21}nh\nu_0}{c\Delta\nu}(N_2 - N_1)$$

$$g_{th} = \alpha_c + \frac{1}{2L} ln\left(\frac{1}{R_1 R_2}\right)$$

$$\eta_{slope} \approx \frac{P_0}{I - I_{th}} \qquad \eta_{EQE} = \frac{P_0/h\nu}{I/e} \qquad \eta_{PCE} = \frac{P_0}{IV_D}$$

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Light Basics

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \qquad \sin(\theta_c) = \frac{n_2}{n_1} \; (n_1 > n_2)$$

Lenses and ABCD Matrices

Refraction at Spherical Surface

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R} \qquad f_o = \frac{n_1}{n_2 - n_1} R \qquad f_i = \frac{n_2}{n_2 - n_1} R$$

Convex $\implies R > 0$ and concave $\implies R < 0$

Thin Lens

$$\frac{1}{s_o} + \frac{1}{s_i} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{f} \qquad f_i = f_o = f$$

$$\frac{y_o}{|y_i|} = \frac{s_o}{s_i} = \frac{s_o - f}{f} = \frac{f}{s_i - f} = \frac{1}{M}$$

ABCD Matrices

Prop.: $\begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}$ Refr.: $\begin{bmatrix} 1 & 0 \\ 0 & n_1/n_2 \end{bmatrix}$ Thin. Lens: $\begin{bmatrix} 1 & 0 \\ -1/f & 1 \end{bmatrix}$

Wave Equation

Plane Wave

$$E_x(z,t) = \Re\left(E_0 e^{j\phi_0} e^{(\omega t - kz)}\right) \quad B_y = \frac{c}{n} E_x \quad \omega = \frac{c}{n} k \quad n = \sqrt{\epsilon_r}$$

$$v_{ph} = \frac{c}{n} \qquad k = \frac{2\pi}{\lambda} = n k_0 \qquad k_0 = \frac{2\pi}{\lambda_0}$$

$$v_g = \frac{c}{n_g} = \left(\frac{dk}{d\omega}\right)^{-1} \qquad n_g = n(\omega) + \omega \frac{dn}{d\omega} = n(\lambda) - \lambda \frac{dn}{d\lambda}$$

Power, Intensity, Irradiance

$$U = U_E + U_M = \epsilon_r \epsilon_0 E_x^2 \qquad U_E = U_M$$

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} \quad S = \frac{1}{\mu_0} E_x B_y = \frac{n}{\eta_0} E_0^2 \sin^2(\omega t) \quad \eta_0 = \sqrt{\mu_0/\epsilon_0}$$

$$I = \langle S \rangle = \frac{1}{2} n \epsilon_0 c |\vec{E}|^2 = \frac{1}{2} \frac{n}{\eta_0} |\vec{E}|^2$$

Polarization

LP.: $\begin{bmatrix} E_{x_o} \\ E_{y_o} \end{bmatrix}$ RCP: $E_o \begin{bmatrix} 1 \\ j \end{bmatrix}$ LCP: $E_o \begin{bmatrix} 1 \\ -j \end{bmatrix}$

Reflection, Refraction, and Fresnel Equations

Fresnel's Equations

$$r_\perp = \frac{E_{r,\perp}}{E_{i,\perp}} = \frac{\cos(\theta_i) - \sqrt{n_r^2 - \sin^2(\theta_i)}}{\cos(\theta_i) + \sqrt{n_r^2 - \sin^2(\theta_i)}}$$

$$t_\perp = \frac{E_{t,\perp}}{E_{i,\perp}} = \frac{2\cos(\theta_i)}{\cos(\theta_i) + \sqrt{n_r^2 - \sin^2(\theta_i)}}$$

$$r_\parallel = \frac{E_{r,\parallel}}{E_{i,\parallel}} = \frac{\sqrt{n_r^2 - \sin^2(\theta_i)} - n_r^2 cos(\theta_i)}{\sqrt{n_r^2 - \sin^2(\theta_i)} + n_r^2 cos(\theta_i)}$$

$$t_\parallel = \frac{E_{t,\parallel}}{E_{i,\parallel}} = \frac{2 n_r \cos(\theta_i)}{\sqrt{n_r^2 - \sin^2(\theta_i)} + n_r^2 cos(\theta_i)}$$

$$n_r = n_2/n_1 \qquad t_\perp - r_\perp = 1 \qquad n_r t_\parallel + r_\parallel = 1$$

$$r_\parallel = 0 \implies \tan(\theta_B) = n_r = \frac{n_2}{n_1}$$

Fresnel's Equations at Normal Incidence

$$r_\perp = r_\parallel = \frac{n_1 - n_2}{n_1 + n_2} \qquad t_\perp = t_\parallel = \frac{2n_1}{n_1 + n_2}$$

Reflectances and Transmittances

$$R_\perp = \frac{I_{r,\perp}}{I_{i,\perp}} = |r_\perp|^2 \qquad R_\parallel = \frac{I_{r,\parallel}}{I_{i,\parallel}} = |r_\parallel|^2$$

At normal incidence:

$$T_\perp = \frac{I_{t,\perp}}{I_{i,\perp}} = \frac{n_2 |E_{t,\perp}|^2}{n_1 |E_{i,\perp}|^2} = \frac{n_2}{n_1} |t_\perp|^2 \qquad T_\parallel = \frac{I_{t,\parallel}}{I_{i,\parallel}} = \frac{n_2}{n_1} |t_\parallel|^2$$

$$R_\perp = R_\parallel = R = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 \qquad T_\perp = T_\parallel = T = \frac{4 n_1 n_2}{(n_1 + n_2)^2}$$

At Oblique Angles:

$$T_\perp = \frac{I_{t,\perp} A_t}{I_{i,\perp} A_i} = \frac{n_2 \cos(\theta_t) |E_{t,\perp}|^2}{n_1 \cos(\theta_i) |E_{i,\perp}|^2} = \frac{n_2 \cos(\theta_t)}{n_1 \cos(\theta_i)} |t_\perp|^2$$

$$T_\parallel = \frac{I_{t,\parallel} A_t}{I_{i,\parallel} A_i} = \frac{n_2 \cos(\theta_t) |E_{t,\parallel}|^2}{n_1 \cos(\theta_i) |E_{i,\parallel}|^2} = \frac{n_2 \cos(\theta_t)}{n_1 \cos(\theta_i)} |t_\parallel|^2$$

Evanescent Waves and TIR

$$k_{t,y} = \pm j n_1 k_o \sqrt{\sin^2(\theta_i) - \left(\frac{n_2}{n_1}\right)^2} = \pm j \alpha_y \qquad \delta = 1/\alpha_y$$

$$\vec{E_t} = \vec{E}_{t,o} e^{j(\omega t - k_{t,x} x)} e^{\alpha y} \qquad \Delta x \approx 2\delta \tan(\theta_i)$$

Interference

$$\vec{E} = \vec{E_1} + \vec{E_2} \implies I = n \epsilon_0 c \langle \vec{E} \cdot \vec{E} \rangle = I_1 + I_2 + 2 n \epsilon_0 c \langle \vec{E_1} \cdot \vec{E_2} \rangle$$

$$I = I_1 + I_2 + I_{12} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\phi_1 - \phi_2)$$

Const. Interference: $\phi_1 - \phi_2 = 2m\pi \qquad I = (\sqrt{I_1} + \sqrt{I_2})^2$

Destr. Interference: $\phi_1 - \phi_2 = (2m+1)\pi \qquad I = (\sqrt{I_1} - \sqrt{I_2})^2$

Interferometers

Mach-Zehnder: $\frac{I_{out}}{I_{in}} = \cos^2(\Delta\phi/2)$

In Michelson: $\Delta\phi = 2 n k_0 (L_2 - L_1)$